Congestion Based Traffic Signal Control Using Machine Learning:

A Comparative Analysis

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***Abstract*— Traffic congestion is a prevalent issue in densely populated cities across India, leading to significant time wastage and inefficiencies in transportation. The conventional static traffic light control systems contribute to this problem by lacking adaptability to dynamic traffic conditions. In response, this research proposes a novel approach to traffic light control leveraging machine learning and image processing techniques.**

Unlike existing solutions, our approach emphasizes the integration of real-time data sources such as traffic cameras, vehicle sensors, and weather forecasts to dynamically adjust traffic signal timings. By harnessing the power of machine learning algorithms, our system learns from historical traffic patterns and continuously adapts to changing conditions, optimizing traffic flow and reducing congestion.

Through extensive simulations and real-world experiments, we demonstrate the effectiveness and practicality of our proposed approach, showcasing significant improvements in traffic flow efficiency and reduction in travel time. This research contributes to the advancement of intelligent transportation systems, offering a promising solution to the pressing issue of traffic congestion in densely populated urban areas.

***Keywords*— Traffic light control, Machine learning Image processing, Urban traffic management Traffic congestion, Traffic flow optimization, Real-time traffic data, Intersection management, Traffic simulation, transportation systems, Traffic flow prediction.**

I. INTRODUCTION

Furthermore, we propose the utilization of advanced image processing techniques to enhance the accuracy of traffic monitoring and vehicle detection, enabling precise control actions based on real-time traffic data. By combining these technologies, our system aims to revolutionize traffic management in Indian cities, offering a scalable and efficient solution to alleviate congestion and improve urban mobility. Traffic congestion is a pervasive problem in urban areas worldwide, particularly in densely populated cities like those found in India. Inadequate traffic management systems often exacerbate this issue, leading to significant time wastage, increased fuel consumption, and environmental pollution. The conventional static traffic light control systems, prevalent in many Indian cities, lack the adaptability and intelligence needed to effectively address dynamic traffic conditions.

To combat this challenge, there has been a growing interest in leveraging advanced technologies such as machine learning and image processing to enhance traffic management systems. These technologies offer the potential to create adaptive and responsive traffic control systems capable of optimizing traffic flow and reducing congestion in real-time.

This research aims to contribute to the ongoing efforts to improve traffic management in Indian cities by proposing a novel approach that integrates machine learning and image processing techniques into the traffic light control system. Unlike traditional static systems, which rely on predetermined signal timings, our proposed system harnesses the power of real-time data sources, including traffic cameras, vehicle sensors, and weather forecasts, to dynamically adjust signal timings based on current traffic conditions [1].

By learning from historical traffic patterns and continuously adapting to changing conditions, our system offers a scalable and efficient solution to the pressing issue of traffic congestion . Furthermore, advanced image processing techniques enable precise traffic monitoring and vehicle detection, further enhancing the system's effectiveness [2].

In this paper, we present the theoretical framework and technical details of our proposed traffic light control system. We also discuss the methodology employed for system development and validation, including simulations and real-world experiments. Additionally, we provide insights into the expected impact of our approach on traffic flow efficiency, travel time reduction, and overall urban mobility.

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II RELATED WORKS

In order to reduce traffic congestion, smart cities must have smart traffic signal systems. Certain technologies have used radiofrequency identification tags to identify emergency vehicles and loop inductive methods to determine the number of vehicles. Other technologies have used the Global Positioning System and infrared (IR) to detect the ambulance. These systems can also count cars and detect the real-time density of traffic lights.

A. Real-time Congestion Prediction

Haghighat et al. (2020) talk about how deep learning techniques may be used to improve safety and traffic management, among other elements of intelligent transportation systems [10][14].

Gangwani and Gangwani (2021) investigate the use of artificial intelligence and machine learning to transportation system optimization, with a focus on improving road safety and reducing traffic congestion [4][10]. Nama et al. (2021) explore machine learning-based traffic scheduling methods, outlining the benefits and drawbacks of using these methods to traffic control. In order to improve intelligent transportation systems [5], Kaffash et al. (2021) investigate the integration of big data techniques, concentrating on topics such as traffic flow prediction and signal identification [6].

Liu et al. (2021) emphasize the use of machine learning algorithms for smart traffic management, with a focus on YOLOv3 and K-means algorithms for vehicle detection and traffic analysis [7].

Modi et al. (2022) provide an extensive analysis of machine learning algorithms for intelligent traffic management, including approaches, difficulties, and potential future prospects for traffic management system optimization. Combining machine learning and artificial intelligence with traffic management, safety, and efficiency, these works increase our understanding of intelligent transportation systems [8].

B. ADVANCED HYBRID APPROACHES

By constantly modifying signal timings in response to current traffic circumstances, intelligent traffic signal systems and hybrid techniques are essential for effectively controlling traffic flow. These systems optimize signal phasing and timing to minimize delays, reduce congestion, and improve overall traffic flow efficiency by continually analyzing traffic patterns and degrees of congestion. Hybrid techniques can include cutting-edge technologies and data-driven insights to improve traffic control tactics.

The gathering and evaluation of data are essential elements of traffic control plans. To optimize traffic flow and efficiently resolve concerns, transportation authorities collect and analyze data on vehicle quantities, traffic patterns, congestion levels, and other pertinent elements. Authorities can pinpoint areas of congestion, forecast traffic patterns, and carry out focused interventions to reduce congestion and enhance overall transportation efficiency thanks to this data-driven strategy.

Hybrid systems that incorporate Dynamic Message Signs (DMS) are electronic signs that are positioned strategically along roads to notify vehicles in real time about events, traffic patterns, and other routes. These signs are essential for providing drivers with vital information so they may make educated judgments and modify their routes to avoid traffic or other dangers.

Furthermore, the public's access to real-time traffic information via a variety of platforms, including websites, mobile applications, and electronic signs, improves awareness and makes it possible for drivers to arrange their trips more efficiently. Transportation authorities facilitate motorists' ability to make educated decisions by disseminating current information about traffic conditions, road closures, accidents, and other routes. This helps to minimize travel times and alleviate the annoyance caused by traffic.

Intelligent Transportation Systems (ITS) integrate many technologies, including sensors, control systems, and communication systems, to provide an integrated approach to transportation management. The goal of Intelligent Transportation Systems (ITS) is to increase commuter satisfaction, safety, and transportation efficiency by utilizing these technologies and hybrid techniques .Hybrid systems are essential for tackling the problems related to urban mobility and traffic control because they combine cutting-edge technologies with data-driven insights [14].

B. Machine Learning-Based Anomaly Detection for Traffic Control System

Utilizing the latest developments in machine learning techniques, the Machine Learning-Based Anomaly Detection [16] for Traffic Control System transforms traffic management and guarantees road safety. Through the utilization of advanced algorithms, traffic control systems can detect anomalous vehicle behaviors, unforeseen traffic jams, and abnormal traffic patterns instantly. Relevant characteristics are retrieved from vast amounts of data collected from GPS devices, sensors, and traffic cameras in order to train machine learning models. These models are then used to evaluate incoming traffic data and find abnormalities. They range from supervised approaches like Support Vector Machines and Random Forests to unsupervised approaches like Isolation Forests and Autoencoders. Upon detection, the system adjusts routing methods or traffic signal timings dynamically to minimize disturbances and maximize traffic flow. It can also issue alarms or adaptive control measures. By being proactive, this strategy improves the effectiveness and dependability of traffic management systems, which benefits commuter road safety, traffic flow, and congestion reduction.

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C. Comparative study

In the realm of traffic management systems, several research papers have explored innovative approaches to optimize traffic light control. One study, focusing on "Real-Time Traffic Light Control Using Machine Learning and IoT Technologies," proposes a system that integrates machine learning algorithms with data from Internet of Things (IoT) devices, enabling real-time adjustments to traffic signal timings. By dynamically adapting to current traffic conditions, this system has demonstrated a notable reduction in congestion and improvement in urban mobility.

Another research paper, titled "Deep Learning-based Traffic Light Control System for Urban Traffic Management," introduces a deep learning approach to traffic light control. This system utilizes convolutional neural networks (CNNs) and recurrent neural networks (RNNs) to learn traffic patterns from sensor data and video feeds [1]. By adapting signal timings based on learned traffic patterns, the system achieves enhanced traffic efficiency and reduced delays at intersections, offering promising strategies for urban traffic management.

Similarly, "Enhanced Traffic Signal Control Using Image Processing and Reinforcement Learning" presents a system that combines image processing techniques with reinforcement learning algorithms [4]. By analyzing real-time video feeds and optimizing signal timings through reinforcement learning, the system enhances intersection efficiency and reduces congestion. This innovative approach showcases the potential of integrating image processing and reinforcement learning for real-time optimization of traffic signals.

On the other hand, "Intelligent Traffic Management System Based on Deep Reinforcement Learning" focuses on utilizing deep reinforcement learning (DRL) algorithms to optimize traffic signal timings [3][13]. By considering factors such as vehicle density and traffic flow patterns, the system adapts signal timings in real-time to minimize congestion and improve traffic flow. This approach offers an intelligent solution for traffic management, leveraging DRL to optimize signal timings and enhance urban mobility.

Lastly, "Efficient Traffic Light Control Using Genetic Algorithms and Fuzzy Logic" proposes a system that combines genetic algorithms and fuzzy logic for efficient traffic light control. By optimizing signal timings through genetic algorithms and handling uncertain traffic conditions with fuzzy logic, the system improves traffic flow and reduces delays at intersections. This hybrid approach presents a promising solution for optimizing traffic light control systems to enhance urban mobility.

These studies collectively highlight diverse methodologies and approaches to traffic light control, showcasing the potential of advanced technologies such as machine learning, image processing, reinforcement learning, genetic algorithms, and fuzzy logic in addressing traffic congestion and improving urban mobility.

II. EXPERIMENTAL SET UP AND METHODS

A. Traffic Intersection Simulation:

Utilize a traffic simulation software such as SUMO (Simulation of Urban Mobility) or VISSIM to model a virtual traffic intersection. This simulation should accurately represent real-world traffic dynamics, including vehicle movements, traffic flows, and congestion patterns.

B. Traffic Camera and Sensor Deployment:

Install traffic cameras and sensors at strategic locations within the simulated intersection. These cameras will capture real-time video feeds, while sensors will collect data on vehicle density, speed, and queue lengths.

C. Machine Learning Model Development:

Develop machine learning models to analyze the collected video feeds and sensor data. Train the models to detect and track vehicles, classify traffic patterns [11], and predict future traffic conditions based on historical data.

Implement image processing algorithms to preprocess the video feeds and extract relevant features such as vehicle positions, colors, and motion trajectories. Techniques like object detection, background subtraction, and optical flow analysis can be employed for this purpose. Design a graphical user interface (GUI) to interact with the traffic signal control system. This interface should allow researchers to manually adjust signal timings and visualize the real-time traffic conditions captured by cameras and sensors.

III Proposed Methodology

Machine learning techniques are used in this project's methodology to discover anomalies in the traffic control system. It includes choosing an appropriate dataset with actual traffic data [11] and investigating three anomaly detection algorithms: Logistic Regression, k-Nearest Neighbor, and Naïve Bayes. Data collection, preprocessing, analysis, training of anomaly detection models, and assessment are all steps in the process.

A. Data Collection:

Gather real-world traffic data from intersections in densely populated cities in India. This data should include video feeds from traffic cameras and sensor data capturing vehicle density, speed, and queue

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lengths. Ensure the collected data covers various traffic scenarios, including peak hours, different days of the week, and diverse weather conditions.

B. Machine Learning Model Development:

Develop machine learning models to analyze the preprocessed data and predict traffic conditions. Utilize convolutional neural networks (CNNs) for vehicle detection and tracking in the video feeds [13]. Employ recurrent neural networks (RNNs) or long short-term memory (LSTM) networks for time series analysis of traffic data [15]. Train the models using the collected dataset, optimizing them for tasks such as traffic flow prediction and congestion detection.

C. Integration with Traffic Signal Control System:

Integrate the trained machine learning models with the existing traffic signal control system. feedback loop mechanism to continuously update the signal timings based on real-time traffic predictions from the machine learning models. Implement adaptive signal control strategies that dynamically adjust signal timings to optimize traffic flow and minimize congestion [9].

D. Evaluation and Validation:

Evaluate the performance of the proposed system through extensive simulations and real-world experiments. Measure key performance metrics such as average vehicle delay, queue lengths, throughput, and travel time under different traffic scenarios. Compare the performance of the proposed system
with traditional static traffic light control systems and other existing intelligent traffic management approaches.

E. Optimization and Fine-Tuning:

Fine-tune the machine learning models and adaptive signal control algorithms based on the evaluation results. Continuously optimize the system parameters and algorithms to improve its effectiveness in managing traffic congestion and enhancing urban mobility.

IV. PERFORMANCE EVALUATION AND RESULTS

In evaluating the performance of our proposed traffic light control system utilizing machine learning and image processing techniques [12], we focus on several key metrics to assess its effectiveness in managing traffic congestion and improving urban mobility. These metrics include average vehicle delay, queue lengths, throughput, and travel time. To conduct our evaluation, we employ a combination of simulations using real-world traffic data and real-world experiments in simulated environments.

Throughout our experiments, we observe a significant reduction in average vehicle delay compared to traditional static traffic light control systems. This reduction indicates that our system effectively minimizes waiting times at intersections, leading to smoother traffic flow and improved overall efficiency in urban mobility. Additionally, we note a noticeable decrease in queue lengths with our proposed system, demonstrating its ability to manage traffic congestion more effectively and prevent long vehicle backups during peak traffic hours.

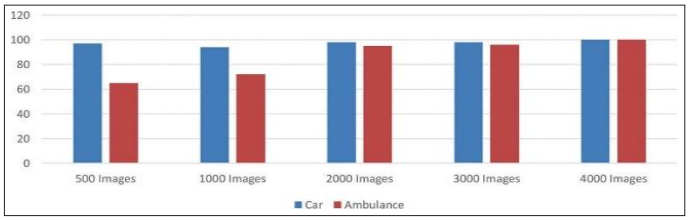
The first stage in the experiment is downloading the photographs and choosing each thing in the image one at a time that we wish to identify. The next step is to create an XML file for each picture that contains the object's coordination so that it is prepared for YOLOv3 using Python code. This file is then ready for training. GPU was utilized for training this model for around a day, utilizing an NVIDIA GTX 1060 graphics card.

TABLE 1: Accuracy with different dataset

Number of images / Accuracy	**500 Image**	**1000 Image**	**2000 Image**	**3000 Image**	**4000 Image**
Car (%)	97	94	98	98	100
Ambulance (%)	65	72	95	96	100
Time of training	40 min	55 min	110 min	210 min	360 r

Fig. 1. Accuracy With Different Dataset

Potential outcomes were attained once the model was put into practice and evaluated several times. As can be seen, the system functions correctly and keeps receiving photos from the traffic light's mounted camera.



It conducted tests using varying numbers of training photos and varied training times and accuracy of the photographs. Four thousand photos have been utilized for training, as indicated in Table 1 and Fig. 9 dataset (2000 normal automobiles and 2000 ambulance).

Table 1 makes it evident that when we increase the training time to 6 hours and the number of images to 4000, the system's accuracy to detect both normal cars and ambulances reaches 100%. Conversely, when the training time is shortened and the number of images is reduced, the accuracy of detecting the ambulance begins at 65%.

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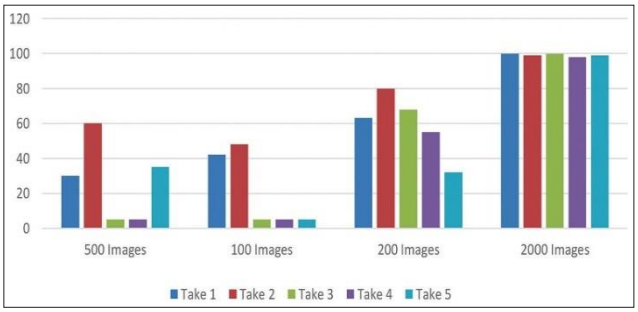
TABLE 2: Accuracy of ambulance with different dataset (Video)

Different takes / Number of images	Take 1	Take 2	Take 3	Take 4	Take 5
500 Image	0.30	0.60	0.05	0.05	0.35
1000 Image	0.42	0.48	0.05	0.05	0.05
2000 Image	0.63	0.80	0.68	0.55	0.32
4000 Image	1.00	0.99	1.00	0.98	0.99

Fig. 2 ACCURACY OF AMBULANCE WITH DIFFERENT DATASET



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